

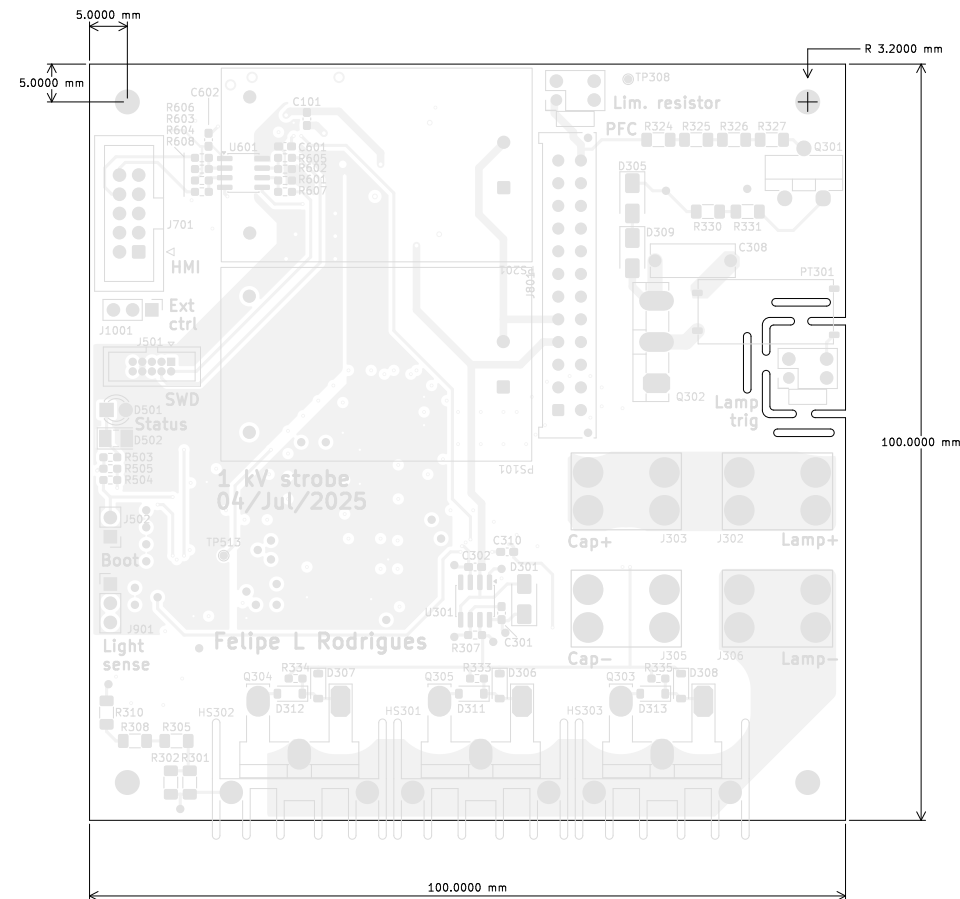
# Main board Fabrication Document

### Top Fabrication (Scale 1:1)

### Layer Stack Legend

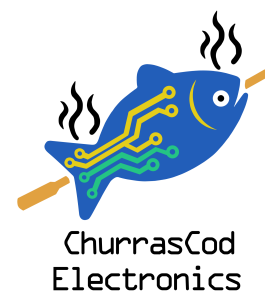
|                                                                                   | Material | Layer             | Thickness        | Dielectric      | Type        | Gerber |
|-----------------------------------------------------------------------------------|----------|-------------------|------------------|-----------------|-------------|--------|
|  |          | F.Paste           |                  |                 | Paste Mask  |        |
|  |          | F.Silkscreen      |                  | Direct Printing | Legend      | GBR    |
|  |          | F.Mask            | 0.02mm           | Solder Resist   | Solder Mask | GBR    |
|  | Copper   | F.Cu (Sig, PWR)   | 0.07mm (2.00oz)  |                 | Signal      | GBR    |
|  | Prepreg  |                   | 0.1mm            | FR4_7628        | Dielectric  |        |
|  | Copper   | In1.Cu (GND)      | 0.035mm (1.00oz) |                 | Plane       | GBR    |
|  | Core     |                   | 1.828mm          | FR4             | Dielectric  |        |
|  | Copper   | In2.Cu (Sig, PWR) | 0.035mm (1.00oz) |                 | Plane       | GBR    |
|  | Prepreg  |                   | 0.1mm            | FR4_7628        | Dielectric  |        |
|  | Copper   | B.Cu (Sig, PWR)   | 0.07mm (2.00oz)  |                 | Signal      | GBR    |
|  |          | B.Mask            | 0.02mm           | Solder Resist   | Solder Mask | GBR    |
|  |          | B.Silkscreen      |                  | Direct Printing | Legend      | GBR    |
|  |          | B.Paste           |                  |                 | Paste Mask  |        |

Total thickness: 2.278mm  
Note: external layer thicknesses are specified after plating



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

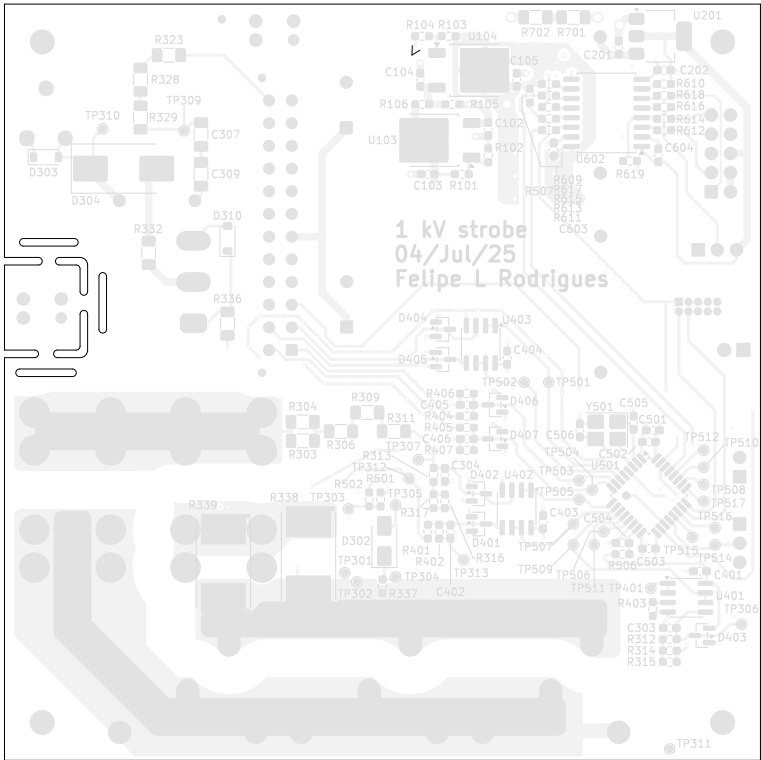
- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge\_Cuts,GBR" SUFFIX.
- DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.
- SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 4) SURFACE FINISH: HAL SNPB
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:
- A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
- B. Tg 170 C OR EQUIVALENT.
- C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY CHURRASCOID ELECTRONICS.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:
- |                  |                    |
|------------------|--------------------|
| BOARD SIZE       | 100.00 × 100.00 mm |
| BOARD THICKNESS  | 2.278 mm           |
| TRACE WIDTH      | 0.300 mm           |
| TRACE TO TRACE   | 0.200 mm           |
| MIN. HOLE (PTH)  | 0.300 mm           |
| MIN. HOLE (NPTH) | 1.000 mm           |
| ANNULAR RING     | 0.100 mm           |
| COPPER TO HOLE   | 0.300 mm           |
| COPPER TO EDGE   | 0.500 mm           |
| HOLE TO HOLE     | 0.250 mm           |

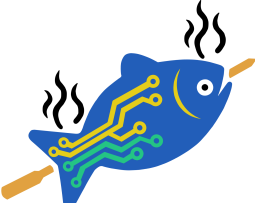


|                                             |                                     |                 |                              |                             |
|---------------------------------------------|-------------------------------------|-----------------|------------------------------|-----------------------------|
| Comments:                                   | Company:<br>ChurrasCod Electronics  |                 | Variant:<br>CHECKED          | Git Hash:<br>a7b8cc5        |
|                                             | Board Name:<br>Main board           |                 | Project Name:<br>1 kV strobe |                             |
| Sheet Title:<br>Top Fabrication (Scale 1:1) | File Name:<br>Strobe_main.kicad_pcb | Designer:<br>FR | Date:<br>2024-04-13          | Revision:<br>+ (Unreleased) |
| Sheet Path:                                 |                                     | Reviewer:       | Size:<br>A3                  | Sheet:<br>1 of 10           |

Main board Fabrication Document

Bottom Fabrication (Scale 1:1)



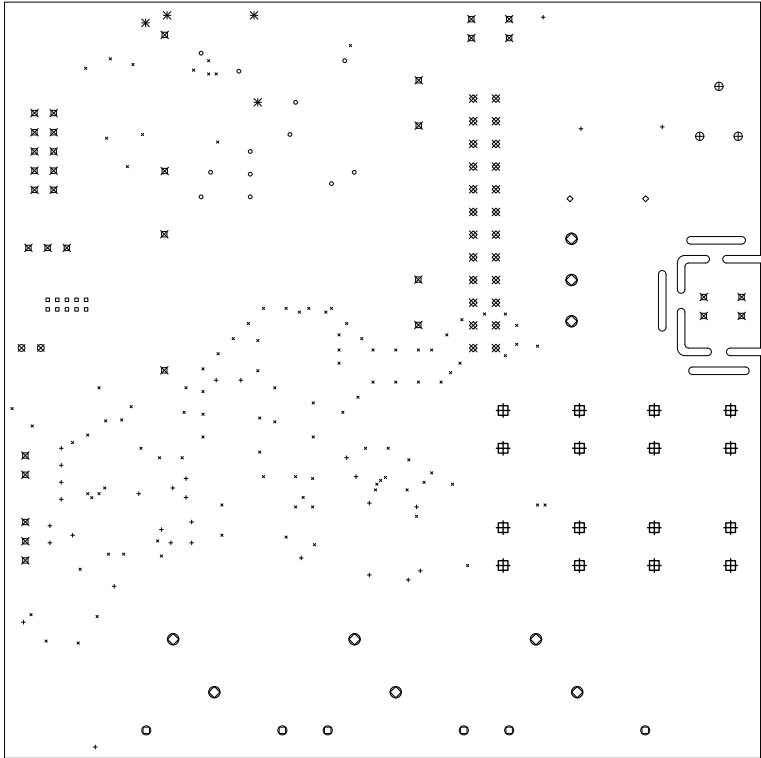
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| <br>ChurrasCod<br>Electronics | Comments:                                      |                                     | Company:<br>ChurrasCod Electronics |                 | Variant:<br>CHECKED          | Git Hash:<br>a7b8ec5        |
|                                                                                                                    |                                                |                                     | Board Name:<br>Main board          |                 | Project Name:<br>1 kV strobe |                             |
|                                                                                                                    | Sheet Title:<br>Bottom Fabrication (Scale 1:1) | File Name:<br>Strobe_main.kicad_pcb |                                    | Designer:<br>FR | Date:<br>2024-04-13          | Revision:<br>+ (Unreleased) |
|                                                                                                                    | Sheet Path:                                    |                                     |                                    | Reviewer:       | Size:<br>A3                  | Sheet:<br>2 of 10           |

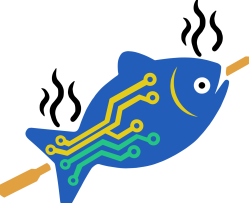
Main board Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

| Symbol | Count     | Hole Size          | Plated | Hole Shape | Drill Layer Pair                  | Hole Type |
|--------|-----------|--------------------|--------|------------|-----------------------------------|-----------|
| ×      | 109       | 0,30mm (11,81mils) | PTH    | Round      | F,Cu (Sig, PWR) - B,Cu (Sig, PWR) | Via       |
| ○      | 12        | 0,50mm (19,69mils) | PTH    | Round      | F,Cu (Sig, PWR) - B,Cu (Sig, PWR) | Via       |
| +      | 31        | 0,50mm (19,69mils) | PTH    | Round      | F,Cu (Sig, PWR) - B,Cu (Sig, PWR) | Pad       |
| □      | 10        | 0,65mm (25,59mils) | PTH    | Round      | F,Cu (Sig, PWR) - B,Cu (Sig, PWR) | Pad       |
| ◇      | 2         | 0,80mm (31,50mils) | PTH    | Round      | F,Cu (Sig, PWR) - B,Cu (Sig, PWR) | Pad       |
| ⊗      | 2         | 0,90mm (35,43mils) | PTH    | Round      | F,Cu (Sig, PWR) - B,Cu (Sig, PWR) | Pad       |
| ※      | 4         | 1,00mm (39,37mils) | PTH    | Round      | F,Cu (Sig, PWR) - B,Cu (Sig, PWR) | Via       |
| ⊠      | 34        | 1,00mm (39,37mils) | PTH    | Round      | F,Cu (Sig, PWR) - B,Cu (Sig, PWR) | Pad       |
| ⊗      | 24        | 1,02mm (40,16mils) | PTH    | Round      | F,Cu (Sig, PWR) - B,Cu (Sig, PWR) | Pad       |
| ⊕      | 3         | 1,10mm (43,31mils) | PTH    | Round      | F,Cu (Sig, PWR) - B,Cu (Sig, PWR) | Pad       |
| ○      | 6         | 1,20mm (47,24mils) | PTH    | Round      | F,Cu (Sig, PWR) - B,Cu (Sig, PWR) | Pad       |
| ○      | 9         | 1,50mm (59,06mils) | PTH    | Round      | F,Cu (Sig, PWR) - B,Cu (Sig, PWR) | Pad       |
| ⊕      | 16        | 1,80mm (70,87mils) | PTH    | Round      | F,Cu (Sig, PWR) - B,Cu (Sig, PWR) | Pad       |
|        | Total 262 |                    |        |            |                                   |           |



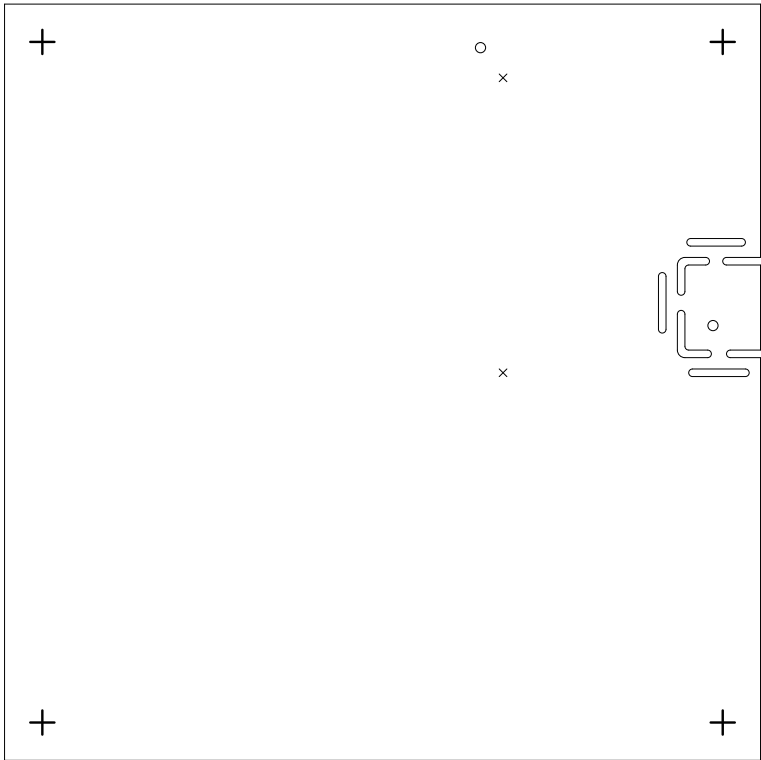
|                                                                                                                 |                                         |                                     |                                    |                     |                              |                      |
|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------|-------------------------------------|------------------------------------|---------------------|------------------------------|----------------------|
| <br>ChurrasCod Electronics | Comments:                               |                                     | Company:<br>ChurrasCod Electronics |                     | Variant:<br>CHECKED          | Git Hash:<br>a7b8ec5 |
|                                                                                                                 |                                         |                                     | Board Name:<br>Main board          |                     | Project Name:<br>1 kV strobe |                      |
|                                                                                                                 | Sheet Title:<br>Drill Drawing (L1 - L4) | File Name:<br>Strobe_main.kicad_pcb | Designer:<br>FR                    | Date:<br>2024-04-13 | Revision:<br>+ (Unreleased)  |                      |
|                                                                                                                 | Sheet Path:                             |                                     | Reviewer:                          | Size:<br>A3         | Sheet:<br>3 of 10            |                      |

Main board Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

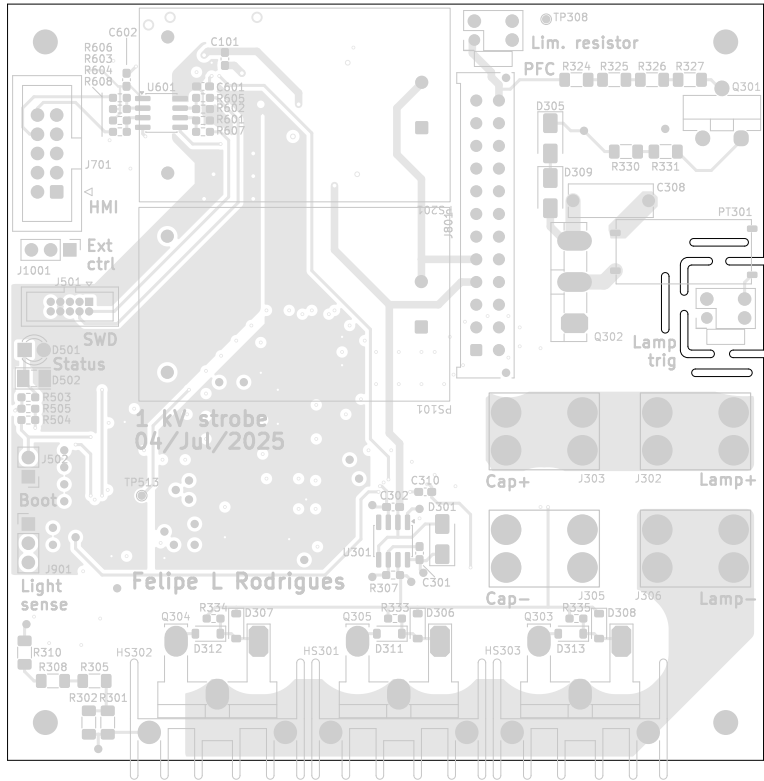
| Symbol  | Count | Hole Size           | Plated | Hole Shape | Drill Layer Pair                  | Hole Type  |
|---------|-------|---------------------|--------|------------|-----------------------------------|------------|
| ×       | 2     | 1,02mm (40,16mils)  | NPTH   | Round      | F.Cu (Sig, PWR) - B.Cu (Sig, PWR) | Mechanical |
| ○       | 2     | 1,35mm (53,15mils)  | NPTH   | Round      | F.Cu (Sig, PWR) - B.Cu (Sig, PWR) | Mechanical |
| +       | 4     | 3,20mm (125,98mils) | NPTH   | Round      | F.Cu (Sig, PWR) - B.Cu (Sig, PWR) | Mechanical |
| Total 8 |       |                     |        |            |                                   |            |



|                                                                                       |                                                |                                            |                                           |                            |                                     |                             |
|---------------------------------------------------------------------------------------|------------------------------------------------|--------------------------------------------|-------------------------------------------|----------------------------|-------------------------------------|-----------------------------|
|  | <b>Comments:</b>                               |                                            | <b>Company:</b><br>ChurrasCod Electronics |                            | <b>Variant:</b><br>CHECKED          | <b>Git Hash:</b><br>a7b8ec5 |
|                                                                                       |                                                |                                            | <b>Board Name:</b><br>Main board          |                            | <b>Project Name:</b><br>1 kV strobe |                             |
|                                                                                       | <b>Sheet Title:</b><br>Drill Drawing (L1 - L4) | <b>File Name:</b><br>Strobe_main.kicad_pcb | <b>Designer:</b><br>FR                    | <b>Date:</b><br>2024-04-13 | <b>Revision:</b><br>+ (Unreleased)  |                             |
| <b>Sheet Path:</b>                                                                    |                                                |                                            | <b>Reviewer:</b>                          | <b>Size:</b><br>A3         | <b>Sheet:</b><br>4 of 10            |                             |

Main board Fabrication Document

Top Test Points (Scale 1:1)



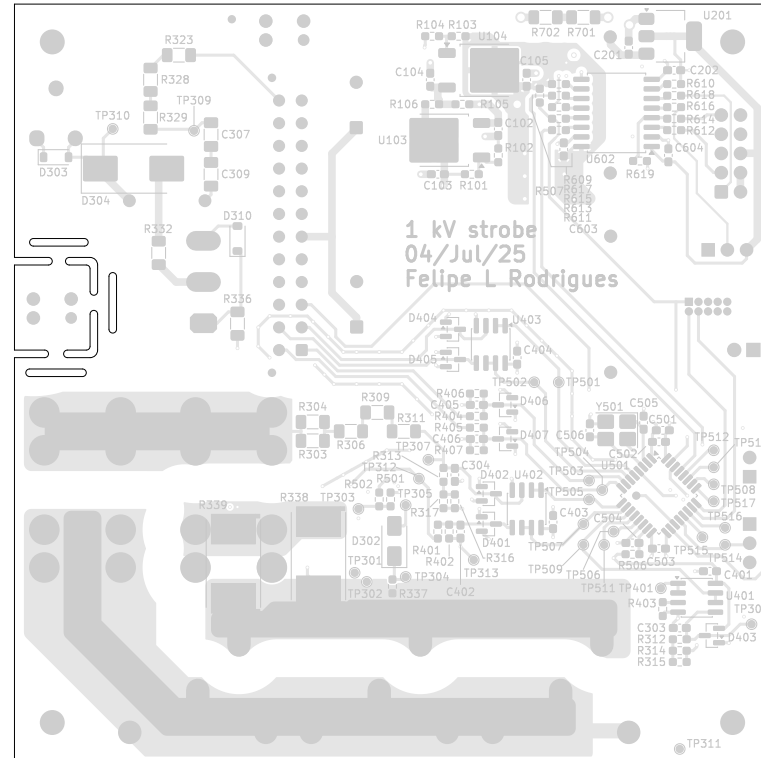
| Ref.  | Net       | X [mm] | Y [mm] |
|-------|-----------|--------|--------|
| TP308 | LAMP_P    | 76.25  | 103.00 |
| TP513 | Vbus_ctrl | 22.75  | 40.00  |



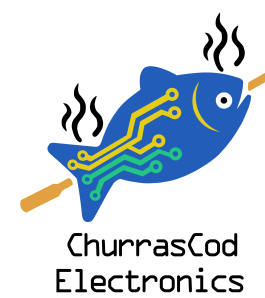
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| Comments:                                   | Company:<br>ChurrasCod Electronics  |                 | Variant:<br>CHECKED          | Git Hash:<br>a7b8ec5        |
|                                             | Board Name:<br>Main board           |                 | Project Name:<br>1 kV strobe |                             |
| Sheet Title:<br>Top Test Points (Scale 1:1) | File Name:<br>Strobe_main.kicad_pcb | Designer:<br>FR | Date:<br>2024-04-13          | Revision:<br>+ (Unreleased) |
| Sheet Path:                                 |                                     | Reviewer:       | Size:<br>A3                  | Sheet:<br>5 of 10           |

# Main board Fabrication Document

### Bottom Test Points (Scale 1:1)



| Ref.  | Net             | X [mm] | Y [mm] |
|-------|-----------------|--------|--------|
| TP301 | VBS             | 60.00  | 29.80  |
| TP302 | DRV_OUT         | 58.40  | 28.60  |
| TP303 | Lamp_EN         | 59.50  | 38.25  |
| TP304 | DRV_RET         | 53.25  | 29.25  |
| TP305 | SH0             | 53.25  | 38.75  |
| TP306 | Um_Col          | 7.50   | 23.00  |
| TP307 | Um_Lmp          | 50.25  | 44.75  |
| TP309 | TP_REF          | 81.25  | 88.25  |
| TP310 | TP_PRI          | 92.00  | 88.50  |
| TP311 | LAMP_N          | 17.00  | 6.50   |
| TP312 | Im_Pulse        | 51.50  | 42.25  |
| TP313 | SH0             | 44.25  | 31.50  |
| TP340 | Free_OUT        | 19.50  | 27.75  |
| TP501 | U_SH1000_buf    | 33.00  | 55.00  |
| TP502 | U_SH240_buf     | 36.25  | 55.00  |
| TP503 | I_SH240         | 29.00  | 42.00  |
| TP504 | I_SH1000        | 27.25  | 40.75  |
| TP505 | U_L_buf         | 29.00  | 39.50  |
| TP506 | Net-(U501-PB0)  | 25.75  | 35.25  |
| TP507 | I_P_buf         | 29.75  | 36.25  |
| TP508 | Net-(U501-PB3)  | 12.50  | 41.50  |
| TP509 | U_C_buf         | 29.75  | 33.50  |
| TP510 | Net-(U501-PB4)  | 12.50  | 43.75  |
| TP511 | Net-(U501-PA7)  | 27.00  | 33.50  |
| TP512 | Net-(U501-PB5)  | 12.50  | 46.00  |
| TP514 | Light_sns       | 11.00  | 33.50  |
| TP515 | INT_EN          | 14.00  | 34.50  |
| TP516 | Net-(U501-PA12) | 11.00  | 35.75  |
| TP517 | Net-(U501-PA15) | 12.50  | 39.25  |



|           |
|-----------|
| Comments: |
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|          |  |
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| Company: |  |
|----------|--|

**ChurrasCod Electronics**

Board Name:

## Main board

|            |  |
|------------|--|
| File Name: |  |
|------------|--|

Strobe\_main.kicad\_pcb

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|          |
|----------|
| Variant: |
|----------|

CHECKED

Git Hash:

a7b8ec5

|               |  |
|---------------|--|
| Project Name: |  |
|---------------|--|

**1 kV strobe**

Date:

2024-04-13

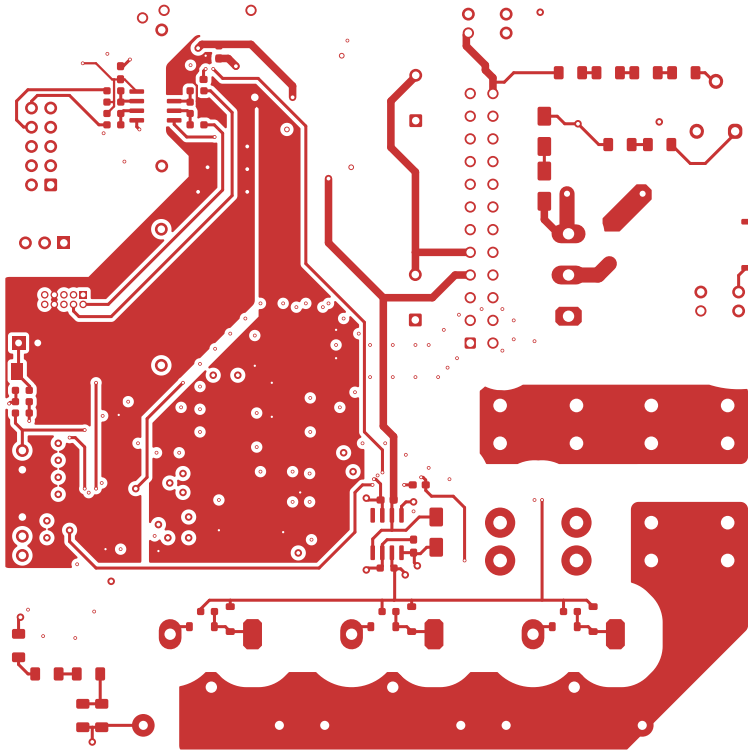
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Main board Fabrication Document

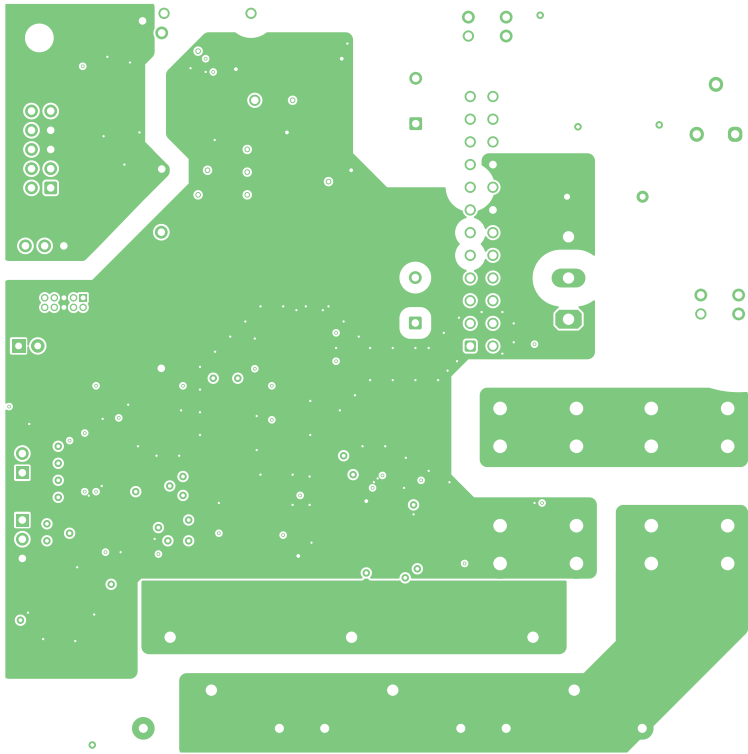
F.Cu (Sig, PWR) (Scale 1:1)

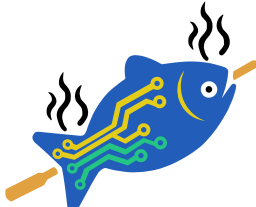


|                                                                                                                    |                                             |                                     |                                    |                 |                              |                             |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------|------------------------------------|-----------------|------------------------------|-----------------------------|
| <br>ChurrasCod<br>Electronics | Comments:                                   |                                     | Company:<br>ChurrasCod Electronics |                 | Variant:<br>CHECKED          | Git Hash:<br>a7b8ec5        |
|                                                                                                                    |                                             |                                     | Board Name:<br>Main board          |                 | Project Name:<br>1 kV strobe |                             |
|                                                                                                                    | Sheet Title:<br>F.Cu (Sig, PWR) (Scale 1:1) | File Name:<br>Strobe_main.kicad_pcb |                                    | Designer:<br>FR | Date:<br>2024-04-13          | Revision:<br>+ (Unreleased) |
|                                                                                                                    | Sheet Path:                                 |                                     |                                    | Reviewer:       | Size:<br>A3                  | Sheet:<br>7 of 10           |

Main board Fabrication Document

In1.Cu (GND) (Scale 1:1)

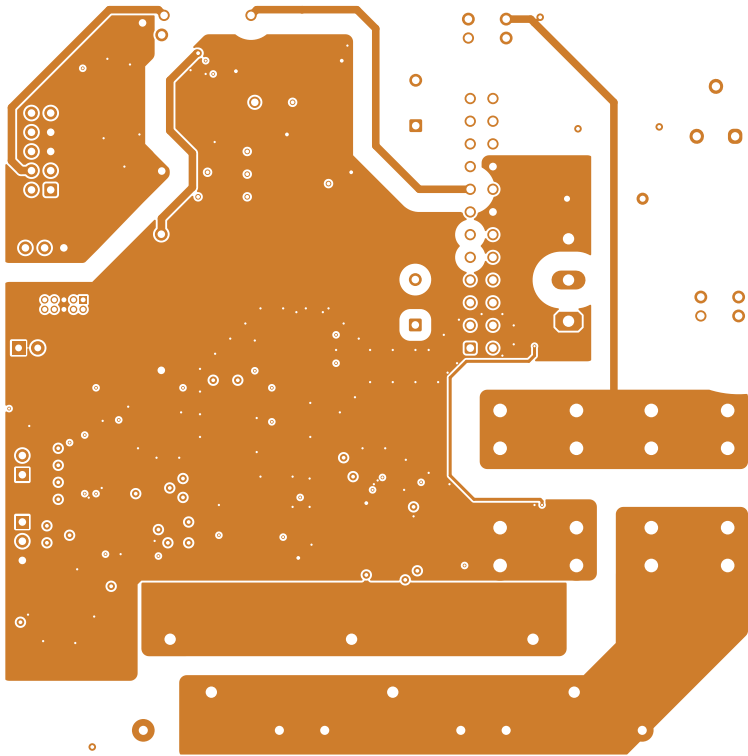


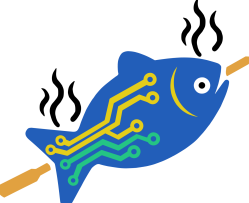
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| <br>ChurrasCod<br>Electronics | Comments:                                | Company:<br>ChurrasCod Electronics  |                 | Variant:<br>CHECKED | Git Hash:<br>a7b8ec5        |
|                                                                                                                    | Sheet Title:<br>In1.Cu (GND) (Scale 1:1) | File Name:<br>Strobe_main.kicad_pcb | Designer:<br>FR | Date:<br>2024-04-13 | Revision:<br>+ (Unreleased) |
|                                                                                                                    | Sheet Path:                              |                                     | Reviewer:       | Size:<br>A3         | Sheet:<br>8 of 10           |



Main board Fabrication Document

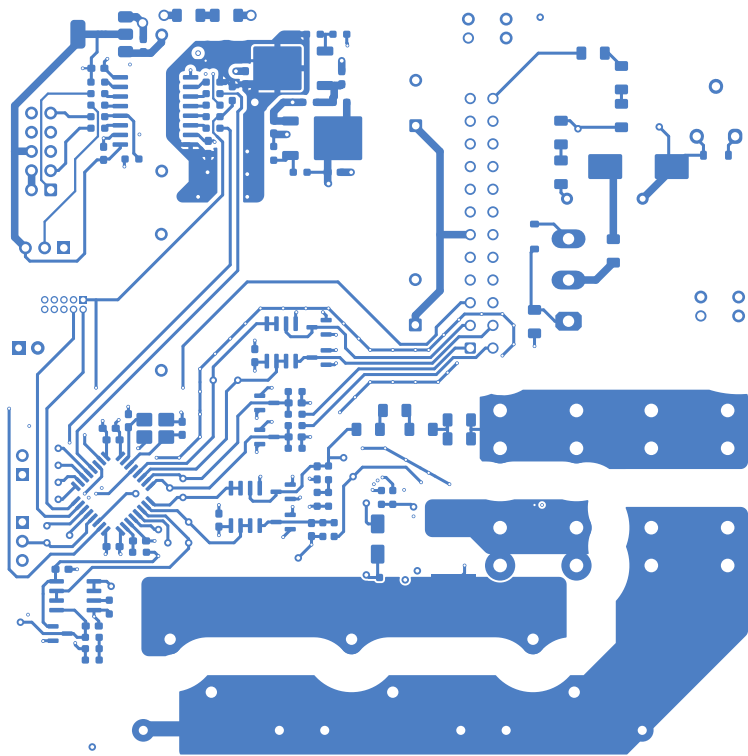
In2.Cu (Sig, PWR) (Scale 1:1)

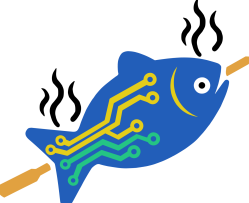


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|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-------------------------------------|-----------------|---------------------|-----------------------------|
| <br>ChurrasCod<br>Electronics | Comments:                                     | Company:<br>ChurrasCod Electronics  |                 | Variant:<br>CHECKED | Git Hash:<br>a7b8ec5        |
|                                                                                                                    | Sheet Title:<br>In2.Cu (Sig, PWR) (Scale 1:1) | File Name:<br>Strobe_main.kicad_pcb | Designer:<br>FR | Date:<br>2024-04-13 | Revision:<br>+ (Unreleased) |
|                                                                                                                    | Sheet Path:                                   | Reviewer:                           |                 | Size:<br>A3         | Sheet:<br>9 of 10           |

Main board Fabrication Document

B.Cu (Sig, PWR) (Scale 1:1)



|                                                                                                                    |                                             |                                     |                 |                              |                             |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------|-----------------|------------------------------|-----------------------------|
| <br>ChurrasCod<br>Electronics | Comments:                                   | Company:<br>ChurrasCod Electronics  |                 | Variant:<br>CHECKED          | Git Hash:<br>a7b8ec5        |
|                                                                                                                    |                                             | Board Name:<br>Main board           |                 | Project Name:<br>1 kV strobe |                             |
|                                                                                                                    | Sheet Title:<br>B.Cu (Sig, PWR) (Scale 1:1) | File Name:<br>Strobe_main.kicad_pcb | Designer:<br>FR | Date:<br>2024-04-13          | Revision:<br>+ (Unreleased) |
|                                                                                                                    | Sheet Path:                                 |                                     | Reviewer:       | Size:<br>A3                  | Sheet:<br>10 of 10          |